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Assessing Axial Load Effects on Downhole Tool Structural Integrity

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Abstract

This paper evaluates the impact of axial forces on drilling tool integrity during drilling, underreaming, and underreaming-only operations. It aims to verify the effects of axial forces on equipment through real runs with the goal of reducing tool damage and enhancing operational efficiency.

The use of under-reamers has become a common approach to address wellbore quality and casing running challenges. This analysis employs detailed 4D simulations of drilling dynamics using real-time data to examine force distributions on the actual Bottom Hole Assembly (BHA). The focus is on lateral and axial vibrations of the cutting structure to determine optimal drilling parameters. Axial forces impacting the tool body within the BHA may be masked by stable parameters, necessitating additional outputs to identify potential axial force fluctuations and guide final drilling parameters.

The paper identified significant concerns arising from the fluctuation of the neutral point during drilling at different Weight on Bit (WOB) and well trajectories, leading to high-frequency axial force variations. This phenomenon can transform the reamer body into a jackhammer-like action, causing high stress concentrations and damage to cutter blocks, the cutting structure, and other internal mechanisms. Over the past five years, statistics show that nearly 17% of Service Quality (SQ) issues are due to underreamer body damage during underreaming operations, making it the second highest cause of tool damage. Understanding axial force distribution dynamics is crucial for minimizing equipment damage and optimizing operational efficiency.

This paper presents novel insights into optimizing reamer placement based on axial force distribution on the BHA. It highlights precautionary measures in parameter selection to enhance drilling dynamics, prevent tool failure, and reduce repair costs.

Introduction

As drilling operations evolve to meet the demands of increasingly complex well architectures, the role of downhole tool reliability has become more critical than ever. Among these tools, underreamers are essential for enlarging boreholes, improving hole quality, and ensuring successful casing and liner deployment.

Despite their importance, underreamers remain vulnerable to mechanical failure, often due to dynamic loads that are not fully captured during the planning phase.

While modern engineering workflows incorporate advanced modeling tools to simulate BHA behavior, these simulations often focus on lateral vibrations, stick-slip, and BHA axial vibrations. Neutral point fluctuation and its effect on BHA dynamics are frequently underestimated or overlooked. These forces, though subtle in surface data, can have a profound impact on tool integrity and lead to significant service quality issues.

This paper was motivated by unexpected tool damage observed during underreaming operations. Despite thorough pre-job planning and dynamic modeling, post-run inspections revealed significant deformation of reamer components. These failures prompted a deeper investigation into the axial dynamics of the BHA, using real-time data and advanced 4D simulation platforms to uncover the hidden forces at play.

The paper not only sheds light on the limitations of conventional modeling approaches but also emphasizes the need for a more nuanced understanding of axial force behavior. By exploring the relationship between neutral point movement, WOB variations, and tool placement, this work aims to enhance the predictive accuracy of pre-job simulations and improve the reliability of downhole tools in demanding drilling environments.

Objectives/Scope

This analysis aims to quantify the impact of axial force fluctuations on underreamer integrity during drilling and underreaming operations in vertical and deviated wells. The specific objectives are:

- To analyze axial force behavior using real-time field specific data
- To investigate the role of the neutral point in concentrating axial loads on the underreamer body, particularly under low WOB and varying trajectory conditions.
- To validate pre-job BHA dynamic models against post-run tool conditions, identifying discrepancies in axial force prediction.
- To perform calibrated 4D simulations to visualize axial load distribution and neutral point movement along the BHA.
- To develop recommendations for BHA design, reamer positioning, and parameter selection to reduce tool damage and improve operational reliability.

Methods, Procedures, and Processes

The process is supported by a comprehensive pre-job engineering analysis. The most geomechanically representative based on formation tops and unconfined compressive strength (UCS) profiles are input and used to simulate the transition from soft to hard formations and assess the expected dynamic behavior of the BHA. As shown in Figure 1, formation logs were used to define the zones of interest and calibrate the dynamic model.

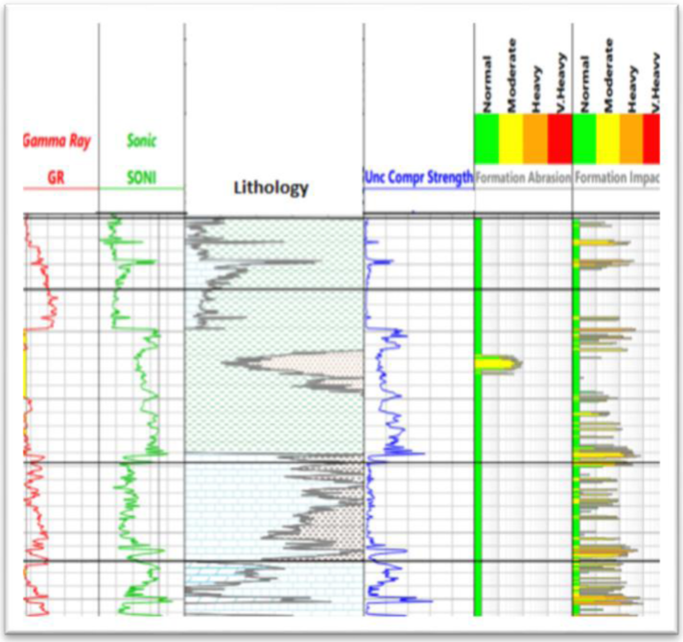


Figure 1—Formation logs

Using modeling simulation software, multiple BHA configurations were simulated to evaluate axial and lateral vibrations, stick-slip tendencies, and the effects of stabilizer placement. The modeling focused on minimizing dynamic shocks and optimizing reamer placement. Several BHA designs were compared under varying WOB and rotating speed conditions. As illustrated in Figure 2, the selected configuration (BHA1) demonstrated minimal axial vibration across all modeled scenarios, with lateral vibration and stick-slip behavior primarily influenced by stabilizer spacing and placement.

		BHA1						BHA2						BHA3						BHA4					
	RPM	60	80	100	120	140	160	60	80	100	120	140	160	60	80	100	120	140	160	60	80	100	120	140	160
	ROP ft/m																								
Axial Vibration (g)	10	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	25	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	40	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.1	0.0	0.0
	55	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.1	0.1	0.0	0.0	0.0	0.1	0.1	0.0	0.0	0.1	0.1	0.1	0.1	0.1
Lateral Vibration (g)	10	0.5	0.6	0.7	0.5	0.5	0.5	0.6	0.5	0.6	0.5	0.6	0.6	0.4	0.9	0.9	0.5	0.4	0.5	0.7	0.4	0.8	0.4	0.5	0.5
	25	1.8	2.0	1.7	1.5	1.3	1.3	5.9	4.1	1.8	1.2	1.1	1.4	1.3	2.8	2.1	1.5	1.5	1.3	2.6	2.2	1.7	0.6	1.0	1.2
	40	3.2	2.4	2.5	2.5	2.3	2.1	1.9	7.9	6.2	4.8	2.4	1.9	2.6	2.5	4.1	4.1	2.5	2.4	1.7	4.4	4.0	3.6	2.6	2.3
	55	0.9	3.8	2.9	3.5	3.3	3.0	0.9	1.8	8.1	8.0	3.0	3.2	1.0	4.8	3.0	7.7	5.3	4.4	0.6	6.2	6.3	5.7	4.1	3.9
S&S (%)	10	123	69	53	77	73	63	130	120	101	108	74	68	128	63	35	68	63	62	128	117	44	107	51	64
	25	69	43	46	30	43	45	33	89	113	118	112	90	81	43	66	81	49	58	82	88	111	149	111	103
	40	44	66	33	20	32	40	67	36	78	74	114	115	47	64	76	70	88	75	75	66	68	47	69	66
	55	88	47	62	21	27	37	86	45	34	44	119	99	82	28	57	33	69	77	91	45	40	35	30	33

Figure 2—Pre-job BHA dynamics simulation

Following the modeling phase, BHA1 was deployed in the field. The operation was executed in a vertical section with known formation transitions. Despite the favorable modeling results, post-run inspection revealed unexpected damage to the underreamer. Deformation was observed on both the upper and lower cutter block areas, as well as on internal mechanical components that interface with the external structure. These signs of high-frequency axial loading are documented in Figures 3 and 4, showing clear evidence of metal deformation and impact damage inconsistent with the predicted stable dynamics.

Figure 3—Metal deformation of external parts.

Figure 4—Metal deformation of internal parts.

To reconcile the discrepancy between the pre-job model and the actual tool condition, a calibrated 4D dynamic simulation was conducted. Real-time field parameters—including WOB, RPM, torque, and rate of penetration were used to replicate the downhole environment. The simulation results, shown in Figure 5, indicate elevated vibration levels when drilling through harder formations. Also, in the collected field data, high vibration was observed, revealing significant fluctuations of the neutral point during drilling, as shown in Figure 6. The neutral point was observed oscillating along the BHA, repeatedly crossing the underreamer body.

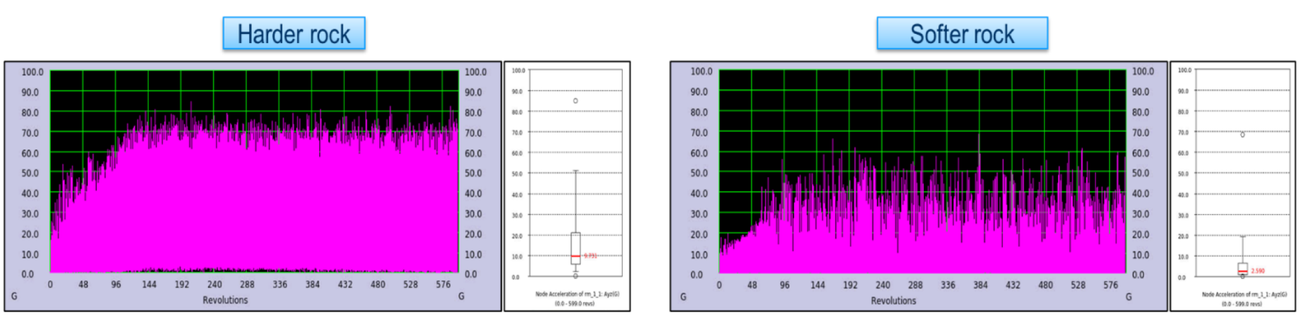


Figure 5—Calibrated lateral vibration levels.

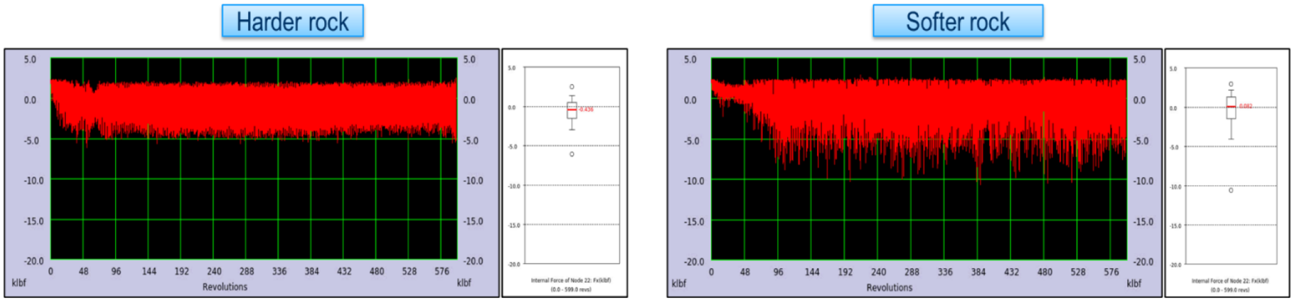


Figure 6—Axial forces on the reamer.

Stress distribution maps from the simulation, presented in Figure 7, highlighted alternating zones of compression and tension concentrated at the reamer location. The animated output further confirmed the presence of high-frequency axial cycling, with the reamer body subjected to jackhammer-like impacts. These findings aligned with the physical damage observed on the tool and provided a clear explanation for the failure mechanism.

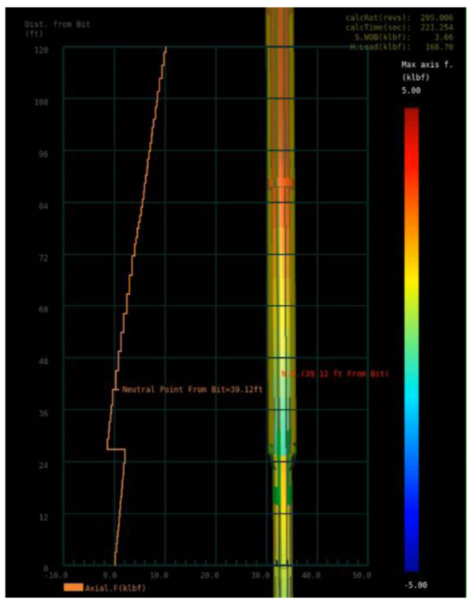
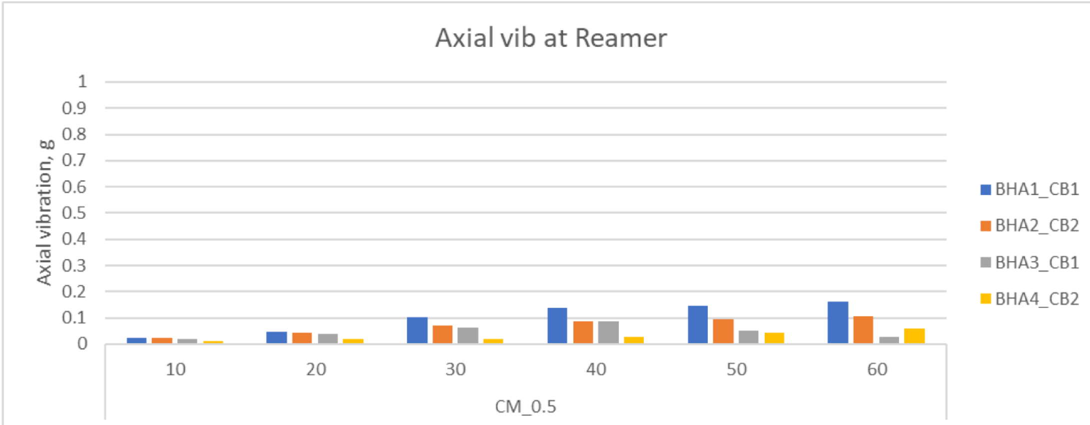
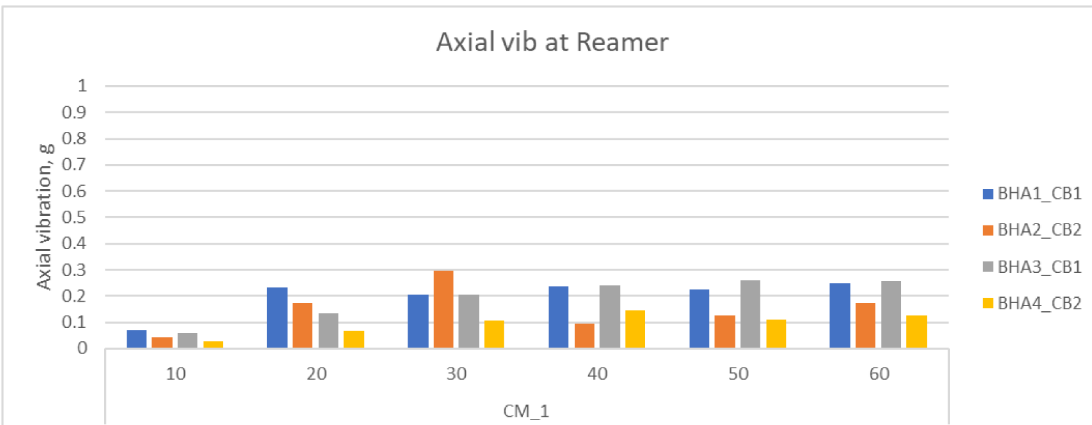


Figure 7—Axial forces on the reamer.

One of the most challenging aspects of the analysis was mitigating tool damage while optimizing service quality and minimizing non-repairable tool costs in future operations. Multiple simulations were conducted to evaluate the impact of BHA stabilization, design modifications, and operational parameters across varying formation tops. Additional shock and vibration analyses were performed to assess dynamic behavior. As illustrated in Graph 1, axial vibration in softer formations showed a correlation with BHA stabilization, increasing with WOB up to a threshold, beyond which it plateaued. A similar simulation using identical BHA design and stabilization was conducted in harder formations, revealing a comparable vibration trend but with higher magnitudes, as shown in Graph 2. Notably, axial vibration levels remained below 0.4G across all scenarios, indicating low vibration intensity. The analysis confirmed consistently low axial vibration on the reamer, independent of cutter block type, BHA configuration, or stabilization strategy. However, low axial vibration observed during pre-job planning may not accurately reflect actual axial loading conditions during execution. This highlights the importance of integrating dynamic modeling with formation-specific parameters to enhance predictive accuracy and tool reliability.



Graph 1—Axial vibration on the reamer in soft formation.

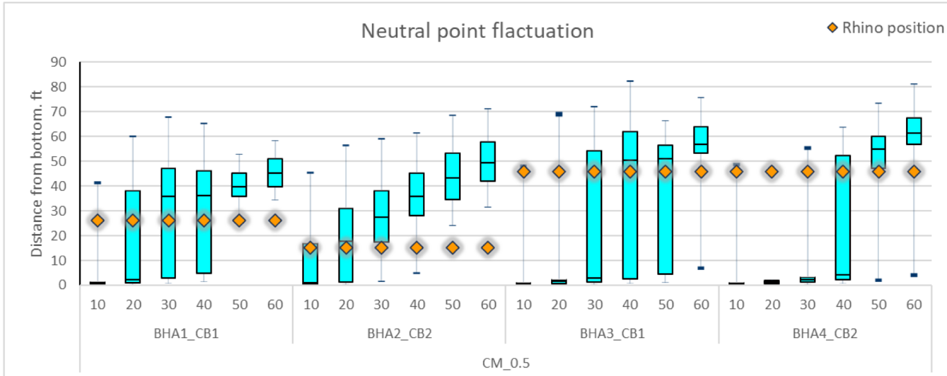


Graph 2—Axial vibration on the reamer in hard formation.

To address neutral point fluctuations and identify critical stress zones along the BHA, a series of simulations were conducted using neutral point fluctuation graphs. These simulations were performed using identical BHA configurations and operational parameters as those used in the axial vibration analysis. The position and stability of the neutral point were found to be highly sensitive to the BHA design, particularly the weight distribution below the reamer.

As illustrated in Graph 3, BHA1 demonstrates that under most operational conditions, the reamer operates within the neutral point zone. This configuration presents a risk of instability due to potential transitions between compression and tension. In contrast, BHA2, which features a reduced weight below the reamer, effectively eliminates the compression-tension crossover zone at lower parameters. However, this configuration requires a higher WOB to maintain the reamer within a stable neutral point range, thereby avoiding excessive fluctuation.

BHA3, similar in design to BHA1, also operates within the critical neutral point region, offering only a narrow operational window for maintaining tool stability. BHA4 follows a similar trend, where increasing drilling parameters is necessary to ensure the reamer remains in a safer, more stable zone.

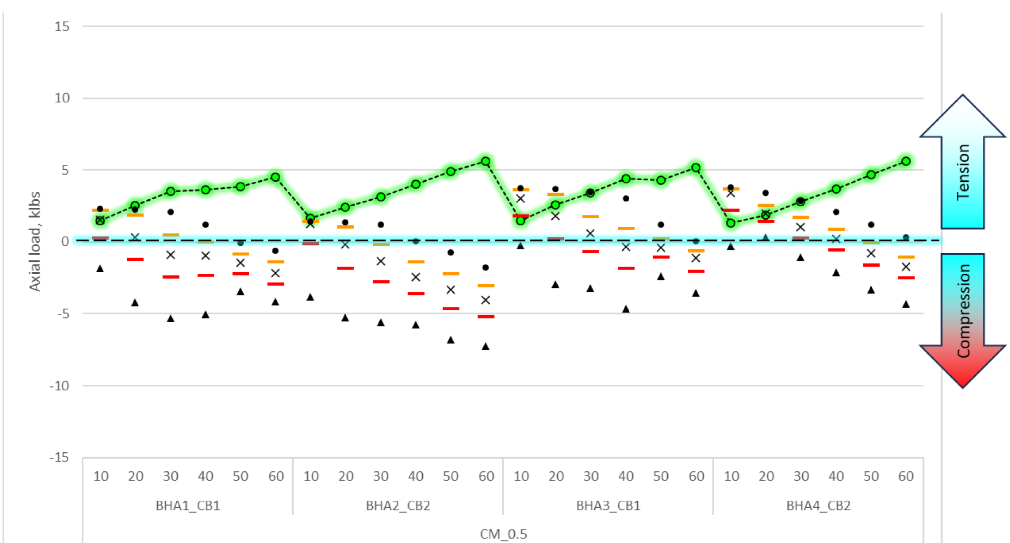


Graph 3—Neutral point fluctuation is based on BHA and parameters.

Overall, the analysis across multiple BHA designs and drilling parameter sets reveals that it is possible to identify optimal configurations—referred to as "sweet spots"—where the reamer consistently operates outside the critical neutral point region. These findings underscore the importance of integrating neutral point fluctuation modeling into pre-job planning to enhance tool reliability and reduce the risk of mechanical failure during drilling operations.

A comprehensive analysis combining axial vibration data with neutral point fluctuation modeling was conducted to accurately visualize axial load distribution along the BHA. This integrated approach enabled precise identification of load magnitudes and locations on critical components, particularly the reamer. The axial load output revealed that during transitions of the neutral point across the tool body, the equipment experiences a dynamic range of loads, fluctuating from negative (compressive) to positive (tension) values, measured in klbs. This cyclical shift between compression and tension modes acts as a jackhammer-like effect, imposing significant mechanical stress on the tool.

As illustrated in Graph 4, BHA1 exhibited axial load variations ranging from −5 klbs (compression) to +3 klbs (overpull) across different operational parameters. These fluctuations are particularly critical during transitions from tension to compression, where the reamer body is subjected to peak stress levels. This phenomenon explains the observed deformation of metallic components under lower WOB conditions, which may initially appear counterintuitive.



Graph 4—Cyclic axial load on the Reamer.

Previous analyses (refer to Graphs 1 and 2) indicated that reducing WOB leads to lower axial vibration levels on the reamer. However, this reduction can be misleading during pre-job planning. Lower vibration readings may suggest safer operating conditions, yet the actual mechanical behavior under reduced WOB can accelerate wear and induce stress concentrations on the reamer body. This misalignment between vibration perception and actual axial loading highlights a critical risk: operating under low parameters may inadvertently increase the likelihood of catastrophic tool failure and compromise service quality.

The analysis underscores the importance of correlating axial vibration data with neutral point fluctuation to obtain a holistic understanding of tool dynamics. By doing so, engineers can identify operational "sweet spots" where the reamer remains in a stable zone, minimizing exposure to harmful load transitions. This insight is vital for optimizing BHA design and selecting appropriate drilling parameters that ensure tool integrity and performance reliability.

The integration of axial load distribution modeling with vibration and neutral point analysis provides a robust framework for predicting tool behavior under varying conditions. It enables proactive mitigation of mechanical risks and supports informed decision-making during both planning and execution phases of drilling operations.

The analysis determined that the primary cause of tool damage was unaccounted axial force fluctuations resulting from instability of the neutral point. These dynamic load variations were not captured during initial modeling due to the assumption of static load distribution and the lack of real-time calibration data. As a result, critical stress cycles acting on downhole tools were overlooked. The analysis further highlighted the consequances of positioning mechanical and electronic tools—such as jars, underreamers, circulation subs, rotary steerable systems (RSS), and measurement-while-drilling (MWD) tools—near the neutral point. These components are particularly vulnerable to alternating compressive and tensile forces, which can induce fatigue and lead to premature failure. This finding underscores the importance of incorporating dynamic modeling of axial loads and neutral point behavior into pre-job planning to ensure accurate risk assessment and enhance tool reliability in complex drilling environments.

Novel/Additive Information

This paper introduces a novel understanding of axial force behavior in drilling BHA's by identifying dynamic neutral point fluctuation as a critical factor contributing to tool damage, particularly in underreamers. Traditional modeling approaches often assume a static or gradually shifting neutral point, which fails to capture the real-time mechanical interactions occurring during drilling. In contrast, this work demonstrates that the neutral point can oscillate rapidly under specific combinations of WOB, trajectory inclination, and BHA configuration. These oscillations result in high-frequency axial load cycling, which imposes alternating compressive and tensile stresses on sensitive components.

The innovation of this paper lies in the integration of real-time field data with calibrated four-dimensional dynamic simulations. Axial force behavior was visualized along the BHA, revealing compression-tension transitions that were not predicted in pre-job static models. These transitions were particularly pronounced in vertical and near-vertical sections with low WOB, where the underreamer was repeatedly subjected to axial stress reversals. This cyclic loading behavior directly correlates with the physical signs of fatigue and deformation observed on retrieved tools.

A significant contribution of this work is the identification of the neutral point fluctuation zone as a high-risk region for tool placement. Mechanical and electronic tools—such as underreamers, jars, circulation subs, RSS, and MWD tools—are especially vulnerable when positioned within this zone. The study recommends avoiding tool placement near the neutral point, particularly in drilling environments characterized by low WOB or unstable borehole trajectories.

This insight challenges conventional BHA design practices, which often prioritize tool spacing and directional control without accounting for dynamic axial load behavior. By incorporating neutral point fluctuation modeling into the BHA design process, engineers can significantly reduce the likelihood of fatigue-induced failures and improve overall tool reliability. The findings advocate for a change in basic assumptions in pre-job planning, emphasizing the need for dynamic, data-driven modeling to enhance operational safety and performance in complex drilling environments.

Additionally, the study proposes enhancements to pre-job planning workflows, including:

- Neutral point fluctuation analysis as a standard checkpoint in BHA modeling.
- Calibration of dynamic models using offset well data and real-time parameters to improve prediction accuracy.
- Reevaluation of tool placement strategies for all components with moving or fatigue-sensitive parts.

These findings are directly applicable to improving tool reliability, reducing service quality incidents, and optimizing drilling performance in both vertical and deviated wells.

Conclusion

This paper establishes that axial force fluctuations—particularly those induced by dynamic neutral point migration—pose a significant threat to the structural integrity of underreamers during drilling and underreaming operations. Despite stable surface parameters and favorable pre-job modeling predictions, post-run tool inspections and real-time downhole data revealed unexpected high-frequency axial loading. These loads resulted in measurable deformation of both external blades and internal mechanical components of the underreamer.

The root cause was identified through calibrated dynamic simulations, which integrated real-time drilling data with advanced mechanical modeling. These simulations demonstrated that the neutral point is not a fixed location along the BHA, but rather a dynamically shifting zone that oscillates in response to changes in WOB, trajectory inclination, and formation variability. As the neutral point repeatedly traversed the underreamer body, the tool was subjected to alternating compressive and tensile forces—effectively creating a jackhammer-like loading pattern.

This cyclic axial stress, often occurring at high frequency, was not captured in conventional pre-job models that assume quasi-static load distributions. The resulting mismatch between predicted and actual tool behavior highlights a critical limitation in standard BHA design methodologies. Specifically, the placement of sensitive mechanical tools within the neutral point fluctuation zone introduces an elevated risk of fatigue-induced failure.

These findings underscore the necessity of incorporating dynamic axial load modeling into the BHA design and planning process. By accounting for real-time neutral point behavior, engineers can better predict tool exposure to damaging stress cycles and optimize tool placement to enhance operational reliability and service life.

Key takeaways from this investigation include:

- The neutral point is a dynamic feature that must be monitored and modeled in real time.
- Placement of mechanical tools near the neutral point should be avoided, especially in vertical wells or low-WOB conditions.
- Pre-job modeling must be calibrated with offset well data and real-time drilling parameters to accurately predict axial dynamics.
- Incorporating neutral point fluctuation analysis into standard engineering workflows can significantly reduce tool failure rates and improve operational efficiency.

Addressing these hidden axial dynamics can enhance tool reliability, reduce non-productive time, and improve the overall success rate of complex drilling operations. The insights from this study are directly applicable to future well planning and tool deployment strategies across a wide range of drilling environments.

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